

Realtek Wi-Fi SDK for Android L 5.x

ver. 1.0.1

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SDK packages

- hardware/realtek/*

Folder to store config files, private code from Realtek.

Introduction

This document provides a simple guide to help engineers to apply Realtek Wi-Fi solution onto their Android L 5.x system. For now, we have supported the following two scenarios:

- **STA/AP** – Switch between STA mode and AP mode
- **(STA+P2P)/AP** – Switch between STA+P2P(Wi-Fi Direct) concurrent mode and AP mode

To port Realtek Wi-Fi driver onto Android 5.x platform, you can go through the following guide with reference codes within our driver package's `realtek_wifi_SDK_for_android_L_5.x_20150811.tgz`.

Because Android's SDK may differ from platform to platform, our reference codes may not be applied on every platform without modifications. You should check if our reference code is suitable for you to use.

In this document, `ANDROID_SDK` is the path of top folder of the target Android SDK; this term is used in the following paragraphs.

1. Copy Necessary Files into SDK

After unzipping `realtek_wifi_SDK_for_android_L_5.x_20150811.tgz`, copy the following folder into `ANDROID_SDK/hardware/` folder:

- `hardware/realtek`

2. Platform Related Files

2.1. BoardConfig.mk

To apply Realtek Wi-Fi solution onto your Android 5.x system, you need to define some compile-time variables in `BoardConfig.mk` of your platform. In general, the `BoardConfig.mk` file is located in:

`ANDROID_SDK /device/<soc_vendor_name>/<board_name>/`

Take TI panda board for example:

`ANDROID_SDK /device/ ti/panda/ BoardConfig.mk`

Please define the following compile-time variables below:

```

BOARD_WIFI_VENDOR := realtek
ifeq ($(BOARD_WIFI_VENDOR), realtek)
    WPA_SUPPLICANT_VERSION := VER_0_8_X
    BOARD_WPA_SUPPLICANT_DRIVER := NL80211
    BOARD_WPA_SUPPLICANT_PRIVATE_LIB := lib_driver_cmd_rtl
    BOARD_HOSTAPD_DRIVER      := NL80211
    BOARD_HOSTAPD_PRIVATE_LIB := lib_driver_cmd_rtl

    BOARD_WLAN_DEVICE := rtl8192cu
    #BOARD_WLAN_DEVICE := rtl8192du
    #BOARD_WLAN_DEVICE := rtl8192ce
    #BOARD_WLAN_DEVICE := rtl8192de
    #BOARD_WLAN_DEVICE := rtl8723as
    #BOARD_WLAN_DEVICE := rtl8723au
    #BOARD_WLAN_DEVICE := rtl8189es
    #BOARD_WLAN_DEVICE := rtl8723bs
    #BOARD_WLAN_DEVICE := rtl8723bu

    WIFI_DRIVER_MODULE_NAME  := "wlan"
    WIFI_DRIVER_MODULE_PATH  := "/system/lib/modules/wlan.ko"
    WIFI_DRIVER_MODULE_ARG   := "ifname=wlan0 if2name=p2p0"

    WIFI_FIRMWARE_LOADER    := "rtw_fwloader"
    WIFI_DRIVER_FW_PATH_STA  := "STA"
    WIFI_DRIVER_FW_PATH_AP   := "AP"
    WIFI_DRIVER_FW_PATH_P2P  := "P2P"
    WIFI_DRIVER_FW_PATH_PARAM := "/dev/null"
endif

```

- **BOARD_WIFI_VENDOR := realtek**

To distinguish the platform Wi-Fi device from products of other vendors, we define variable BOARD_WIFI_VENDOR as realtek. This is for compile-time choices to be applied for Realtek Wi-Fi solutions.

- **WPA_SUPPLICANT_VERSION := VER_0_8_X**

For Android L, please set WPA_SUPPLICANT_VERSION as VER_0_8_X to

use wpa_supplicant_8.

- **BOARD_WPA_SUPPLICANT_DRIVER := NL80211**
- **BOARD_WPA_SUPPLICANT_PRIVATE_LIB := lib_driver_cmd_rtl**
- **BOARD_HOSTAPD_DRIVER := NL80211**
- **BOARD_HOSTAPD_PRIVATE_LIB := lib_driver_cmd_rtl**

We use NL80211 as the driver interface for wpa_supplicant and hostapd to communicate with driver and provide lib_driver_cmd_rtl as the private library.

- **BOARD_WLAN_DEVICE**

Realtek provide a variety of Wi-Fi solutions to choose. For now, BOARD_WLAN_DEVICE is not used for any purpose but we suggest setting this variable for your Wi-Fi solution you used.

- **WIFI_DRIVER_MODULE_NAME**
- **WIFI_DRIVER_MODULE_PATH**
- **WIFI_DRIVER_MODULE_ARG**

These three variables will be used in libhardware_legacy (wifi.c) to do insmod and rmmod. The value of WIFI_DRIVER_MODULE_NAME should match the value of MODULE_NAME specified in our driver's Makefile at compile-time. Please refer to "Platform Setting Section in Detail" of:

document/Quick_Start_Guide_for_Driver_Compilation_and_Installation.pdf

- **WIFI_FIRMWARE_LOADER := rtw_fwloader**

This variable will be used in libhardware_legacy (wifi.c) as the name of Wi-Fi firmware loader, which will be executed after driver's insmod and before the executing of wpa_supplicant and hostapd. Setting it to "rtw_fwloader" for calling service rtw_fwloader which provided by Realtek.

- **WIFI_DRIVER_FW_PATH_STA := STA**
- **WIFI_DRIVER_FW_PATH_AP := AP**
- **WIFI_DRIVER_FW_PATH_P2P := P2P**
- **WIFI_DRIVER_FW_PATH_PARAM := /dev/null**

Realtek driver has FW embedded inside, and will automatically load FW at NIC initialization process. Setting these four variables is just to fit the requirement of the libhardware_legacy (wifi.c).

2.2. init.xxx.rc

For Wi-Fi to operate properly, we need some daemons to be defined as service inside init.xxx.rc. In general, the init.xxx.rc file is located in:

ANDROID_SDK/device/<soc_vendor_name>/<board_name>/

Take TI panda board for example:

ANDROID_SDK/device/ti/panda/init.omap4pandaboard.rc.

Please add the service definitions below:

- **rtw_fwloader**

```
service rtw_fwloader /system/bin/rtw_fwloader
    class main
    disabled
    oneshot
```

- **wpa_supplicant**

```
service p2p_supplicant /system/bin/wpa_supplicant \
    -ip2p0 -Dnl80211 -c/data/misc/wifi/p2p_supplicant.conf \
    -e/data/misc/wifi/entropy.bin -N \
    -iwlan0 -Dnl80211 -c/data/misc/wifi/wpa_supplicant.conf \
    -O/data/misc/wifi/sockets \
    -g@android:wpa_wlan0
    class main
    socket wpa_wlan0 dgram 660 wifi wifi
    disabled
    oneshot

service wpa_supplicant /system/bin/wpa_supplicant \
    -iwlan0 -Dnl80211 -c/data/misc/wifi/wpa_supplicant.conf \
    -O/data/misc/wifi/sockets \
    -e/data/misc/wifi/entropy.bin \
    -g@android:wpa_wlan0
    class main
    socket wpa_wlan0 dgram 660 wifi wifi
    disabled
    oneshot
```

- **dhcpcd**

```
service dhcpcd_wlan0 /system/bin/dhcpcd -aABDKL
    class main
    disabled
    oneshot

service dhcpcd_p2p /system/bin/dhcpcd -aABKL
    class main
    disabled
    oneshot

service iprenew_wlan0 /system/bin/dhcpcd -n
    class main
    disabled
    oneshot

service iprenew_p2p /system/bin/dhcpcd -n
    class main
    disabled
    oneshot
```

2.3. Others

For topics mentioned here, you can add the following code segments in any .mk file which your platform will use. Take TI panda board for example:

ANDROID_SDK /device/ ti/panda/device.mk.

- **Add android.hardware.wifi.xml**

To claim Wi-Fi support for your device, please add the rule in the PRODUCT_COPY_FILES variable to copy the permission definition file of Wi-Fi to the /system/etc/permissions/ folder of your system image.

```
PRODUCT_COPY_FILES += \
frameworks/native/data/etc/android.hardware.wifi.xml:system/etc/permissions/android.hardware.
wifi.xml
```

- **Add android.hardware.wifi.direct.xml**

To claim Wi-Fi Direct (P2P) support for your device, please add the rule in the

PRODUCT_COPY_FILES variable to copy the permission definition file of Wi-Fi Direct to the /system/etc/permissions/ folder of your system image.

```
PRODUCT_COPY_FILES += \
frameworks/native/data/etc/android.hardware.wifi.direct.xml:system/etc/permissions/android.hardware.wifi.direct.xml
```

Make sure your driver is configured for STA+P2P concurrent mode or you may encounter error when you open the Wi-Fi. Please refer to **5. Driver Configurations for Android 5.x**

- **Set wifi.interface**

To specify the wifi interface name in Android, a system property named “wifi.interface” is used. For Realtek Wi-Fi driver, Wi-Fi interface name is assigned with “wlan%d”. In general, you should set wifi.interface as “wlan0”.

```
PRODUCT_PROPERTY_OVERRIDES += \
    wifi.interface=wlan0
```

- **Include rtw_fwloader**

To include rtw_fwloader in the system image, add rtw_fwloader into the PRODUCT_PACKAGES variable.

```
ifeq ($(BOARD_WIFI_VENDOR), realtek)
PRODUCT_PACKAGES += rtw_fwloader
#endif
```

3. System Resource Configurations

You should set the following four resource configurations for your platform to configure the network function and enable the corresponding UI interface. In general, you can set the following configurations in your platform dependent config.xml file. Take TI panda board for example:

ANDROID_SDK/device/ti/panda/overlay/frameworks/base/core/res/res/values/config.xml

Or the global config.xml file:

ANDROID_SDK/frameworks/base/core/res/res/values/config.xml

- **networkAttributes**

To define the system’s available network interfaces, make sure the wifi interface

To know the definition and set other upstream connection types, please refer to ANDROID_SDK/frameworks/base/core/java/android/net/ConnectivityManager.java.

- **config_enableWifiDisplay**

To enable Wi-Fi Display(Miracast) function, set config_enableWifiDisplay to true:

```
<bool name="config_enableWifiDisplay">true</bool>
```

4. wpa_supplicant_8

We provide wpa_supplicant_8_L_5.x_rtw_r14967.20150811.tar.gz or newer version in the wpa_supplicant_hostapd/ of our SW release package. You can:

- **Use the wpa_supplicant_8_L_5.x_rtw instead of the original**

1. Backup and remove the original external/wpa_supplicant_8/ folder
2. Extract and copy the wpa_supplicant_8_L_5.x_rtw tar file to the external/ folder of your Android SDK.
3. Rename wpa_supplicant_8_L_5.x_rtw as wpa_supplicant_8.

※ We have enabled the two macros ANDROID_P2P and REALTEK_WIFI_VENDOR by default.

5. Driver Configurations for Android 5.x

Android 5.x support two scenarios for Wi-Fi solution:

- **STA/AP Switch between STA and AP mode**
- **(STA+P2P)/AP Switch between STA+P2P concurrent and AP mode**

The configuration of driver to fit the requirement of each scenario, see the following table:

MACRO	STA /AP	(STA+P2P)/AP	Kernel ver.
CONFIG_IOCTL_CFG80211	Defined	Defined	ver. >= 2.6.35
RTW_USE_CFG80211_STA_EVENT	Defined	Defined	ver. >= 3.2.0
CONFIG_RADIO_WORK	Defined	Defined	-
CONFIG_CONCURRENT_MODE	Undefined	Defined	-
RTW_ENABLE_WIFI_CONTROL_FUNC	Defined for platform device/driver mechanism		

- **CONFIG_IOCTL_CFG80211** is used for driver to enable cfg80211 ioctl

interface, which is required by Realtek Wi-Fi to operate on Android 5.x system.

- **RTW_USE_CFG80211_STA_EVENT** is used for driver to indicate new cfg80211 STA event, which is required by wpa_supplicant_8 of Android 5.x. Linux kernel supports this feature after kernel 3.2. For kernel version between 3.0 and 3.2, please refer to the patch file:

linux-3.0.42_STATION_INFO_ASSOC_REQ_IES.diff

CONFIG_RADIO_WORK is used for driver to fit 'radio work' mechanism of Android 5.x's wpa_supplicant_8. If this MACRO doesn't exist in driver's source code, please contact with Realtek technical windows for suitable driver.

- **CONFIG_CONCURRENT_MODE** is used for driver to enable concurrent mode, which is required by STA+P2P concurrent mode of Android 5.x.

- **RTW_ENABLE_WIFI_CONTROL_FUNC** is used to register platform driver callbacks. If your platform needs those callbacks, please define this macro to register platform driver callback functions. For example, these functions include:

```
static struct platform_driver wifi_device = {  
    .probe          = wifi_probe,  
    .remove         = wifi_remove,
```

By default, the probe callback is used to set up Wi-Fi power and remove callback is used to close Wi-Fi power.

To compile Realtek Wi-Fi driver with the above setting, please refer to the following document:

document/Quick_Start_Guide_for_Driver_Compilation_and_Installation.pdf

Adding platform selection and setting sections for compilation settings of your platform.

For example, if you want to configure Realtek Wi-Fi driver for the (STA+P2P)/AP scenario, make sure the macros: **CONFIG_IOCTL_CFG80211**, **RTW_USE_CFG80211_STA_EVENT**, **CONFIG_RADIO_WORK** and **CONFIG_CONCURRENT_MODE** are defined into the **EXTRA_CFLAGS** settings as following:

```

CONFIG_PLATFORM_ANDROID_L50_SAMPLE = y
...
...
...
ifeq ($(CONFIG_PLATFORM_ANDROID_L50_SAMPLE), y)
EXTRA_CFLAGS += -DCONFIG_LITTLE_ENDIAN
EXTRA_CFLAGS += -DCONFIG_CONCURRENT_MODE
EXTRA_CFLAGS += -DCONFIG_IOCTL_CFG80211 -DRTW_USE_CFG80211_STA_EVENT
EXTRA_CFLAGS += -DCONFIG_RADIO_WORK
ARCH := arm
CROSS_COMPILE := /toolchain/bin/arm-none-linux-gnueabi-
KSRC := / android_sdk/android_l/ kernel
endif

```

6. FAQ

6.1. Wi-Fi (STA mode)

6.1.1. Why Wi-Fi can't enable?

The whole Wi-Fi enabling procedure includes the following three main check points. Please check in sequence:

- **Is network interface(s) created?**
 - insmod driver success
 - Wi-Fi device is recognized
 - wlan0 (and p2p0) is created
- **Does wpa_supplicant run successfully?**
 - wpa_supplicant.conf (and p2p_supplicant.conf) exists and is correct
 - Service definition of wpa_supplicant exists and is correct
 - Binary file wpa_supplicant exists and is executable
- **Do connections of communication socket setup?**
 - Make sure the communication socket settings is matched below:
 - ◆ ctrl_interface in:

```

/data/misc/wifi/wpa_supplicant.conf
(and /data/misc/wifi/p2p_supplicant.conf)

```
 - ◆ Service definition of wpa_supplicant
 - ◆ Paths of communication socket in wifi.c

6.2. Portable Wi-Fi hotspot (AP mode)

6.2.1. Why Portable Wi-Fi hotspot can't enable?

The whole Portable Wi-Fi hotspot enabling procedure includes the following three main check points. Please check in sequence:

- **Is network interface created?**
 - insmod driver success
 - Wi-Fi device is recognized
 - wlan0 is created
- **Does netd and hostapd run successfully?**
 - /data/misc/wifi/hostapd.conf exists and is correct
 - Binary file netd and hostapd exist and are executable
- **Does dnsmasq run successfully?**
 - Binary file dnsmasq exist and are executable

6.3. Wi-Fi Direct (P2P mode)

6.3.1. There is no Wi-Fi Direct UI shown?

Please refer to “**Add android.hardware.wifi.direct.xml**” in chapter 2.3. Others to enable Wi-Fi Direct functionality of Android L.

6.3.2. Wi-Fi Direct can't scan any peer?

First, make sure you have workable Wi-Fi Direct device nearby. Make them into Wi-Fi Direct scanning state. Push “SEARCH FOR DEVICES” button also in our device and wait for a while.

If there is still no peer shown the problem is usually caused by wrong service definition of wpa_supplicant services. Please refer to “**wpa_supplicant**” in chapter 2.2. **init.xxx.rc** to check your service definition of wpa_supplicant.